



Project Proposal

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Research Question & Motivation

Research Question: Do linguistic features in housing listing descriptions influence housing prices beyond structural and location characteristics?

Why This Matters

- Housing is a major financial decision
- Listings include both:
 - objective features (size, rooms, location)
 - subjective descriptions (marketing language)
- It is unclear whether:
 - prices reflect true value
 - or are influenced by how homes are described

Key Idea: Test whether language provides additional explanatory power in predicting housing prices.

Methodology & Statistical Inference

Project Pipeline

01

Data Collection

- Web Scraping
- Housing listings dataset (price, sqft, beds, location)
- Listing descriptions (text data)

02

Feature Engineering

- Convert text into variables:
- description length
 - keyword indicators ("luxury", "modern", etc.)
 - sentiment score

Statistical Modeling

Baseline Model:

- Structural Features only

Full Model:

- Structural + Text Features

Methods from Class

- Linear Regression
- Regression Inference
 - test significance of text features (p-values, CIs)
- ANOVA for Regression
 - compare baseline vs full model

03

04